Jiayu Tang

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EDUCATION

08/2024 – 07/2026 **University of Nebraska-Lincoln, United States**
M.Sc., Food Science and Technology (Major), Statistics (Minor) **GPA: 4.0/4.0**

09/2020 - 06/2024
B.Sc. (Dual Degree), Food Science and Technology **University of Nebraska-Lincoln, United States**
B.Eng. (Dual Degree), Food Science and Engineering **Northwest A&F University, China**
GPA: 3.5/4.0

PUBLICATIONS

In Preparation:

1. **Tang J**[#], Adinchezor C[#], Grimaldo C, Zhang Y, Iwuamadi I, Rasmussen C, Rose D, Majumder K, Deehan E[*]. Exploring the Impact of Different Dietary Pulse Market Classes on Human Gut Microbiota and Health.
2. **Tang J**[#], Rasmussen C[#], Deehan E[*]. Effect of distinct dietary pulses market classes on gut microbiome and health markers: A systematic review of intervention trials.
3. Grimaldo C**, Tang J,** Whisenhunt L, Chai W, Deehan E[*]. Evaluating interrelationships between body composition and immunometabolic dysfunction using bioelectrical impedance analysis: An observational study.

Published:

1. Xiao J, Li Y, Niu L, Chen R, **Tang J**, Tong Z, Xiao C[*]. Effect of adding fermented proso millet bran dietary fiber on micro-structural, physicochemical, and digestive properties of gluten-free proso millet-based dough and cake. *Foods*. 2023; 12(15): 2964.
2. Deng X, Niu L, Xiao J, Guo Q, Liang J, **Tang J**, Liu X, Xiao C[*]. Involvement of intestinal flora and miRNA into the mechanism of coarse grains improving type 2 diabetes: an overview. *Journal of the Science of Food and Agriculture*. 2023; 103(9): 4257-4267.

[#]: co-first authors
[*]: corresponding authors

RESEARCH EXPERIENCE

10/2026 - Present **Department of Infectious Diseases and Immunology**
OPS Research Assistant (Full-time) **College of Veterinary Medicine, University of Florida**
Supervisor: Dr. Yiquan Wang

- Investigating how human antibody framework architecture influences antigen-binding prediction under fixed CDR, antigen, and epitope controls.
- Built and analyzed an in-silico antibody-design pipeline using framework grafting, native controls, and epitope-masked structural evaluation.
- Ran large-scale AlphaFold/ColabFold-based antibody–antigen predictions and evaluated model confidence, docking accuracy, and interface contacts.
- Benchmarked emerging antibody-design tools, including Germinal and RFantibody.

07/2024 – 07/2026 **Department of Food Science and Technology**
Graduate Research Assistant **University of Nebraska-Lincoln**
Advisor: Dr. Edward Deehan

Human Clinical Trials:

- Coordinated a 12-week randomized, crossover clinical trial investigating dry bean market-class–specific effects on gut microbiota and cardiometabolic health markers (IRB Approved; ClinicalTrials.gov Registered)

- Co-developed a NIH-style study protocol, informed consent form, and recruitment materials with the PI; authored 4 Standard Operating Procedures (SOPs) and 2 clinical trial workflows for the Deehan Lab.
- Coordinated a human biobank study aims to collect and store blood and stool samples from volunteers along with their health information (IRB Approved).
- Led 200+ clinic visits, managed participant recruitment, scheduling, documentation, and study supplies.
- Collected and processed stool samples, blood samples, 24-hr blood pressure data, and other dietary information, and applied data formatting, cleaning, and statistical analysis with python and R.
- Performed wet-lab processing of human blood (centrifugation, plasma/serum separation, aliquoting) and stool (homogenization, aliquoting) samples for −80 °C biobanking, and conducted PCR.

Systematic Review:

- Applied Large Language Model (Gemini 3) to assist with preliminary data extraction and formatting, and validated model performance with human reviewers.
- Conducted a systematic review aims to identify the effects of distinct dietary pulse market classes on gut microbiome and surrogate health outcomes in humans.
- Developed a PICOS-based research proposal, conducted systematic literature searching, and applied machine learning-based tools (ASReview, TERA-Tool) for 1,800+ reference data extraction.

04/2022 - 06/2023 **College of Food Science and Engineering**
Undergraduate Research Assistant **Northwest A&F University**
Advisor: Dr. Chunxia Xiao

- Conducted in vitro fermentation experiments to evaluate how three lactic acid bacteria species improve the phenolic content, antioxidant activity, and starch digestibility of *proso* millet powder.
- Performed spectrophotometric analyses to quantify the color characteristics of gluten-free cake crumb and crust made with *proso* millet bran.
- Conducted targeted literature research and summarize key findings on coarse grain consumption across 50+ articles, contributed to a review article determining the relationship between coarse grain consumption and Type 2 diabetes.

TEACHING EXPERIENCE

01/2026 – 05/2026 **Department of Food Science and Technology**
Teaching Assistant **University of Nebraska-Lincoln**

- Prepared experimental materials for FDST 458-153 and led lab sessions.
- Graded assignments for FDST 458.
- Participated in the proctoring and marking of exams for 35 students.

02/2025 – 04/2025 **Department of Food Science and Technology**
Guest Lecturer **University of Nebraska-Lincoln**

- Delivered 3 lectures for FDST 498 on systematic reviews, covering database search strategies, PICOS-based study design, and the application of AI/ML tools to accelerate literature screening and data extraction.

08/2023 - 05/2024 **Department of Food Science and Technology**
Teaching Assistant **University of Nebraska-Lincoln**

- Graded assignments for FDST 412 and FDST 429.
- Graded assignments for NUTR 250.
- Participated in the proctoring and marking of exams for 55 students.

09/2022 - 03/2023 **Foreign Department**
Teaching Assistant (Internship) **Xi'An New Oriental Education & Technology Group**

- Organized and proctored an IELTS practice exam for 71 students.
- Supported academic planning and language exam preparation for 4 students (Z Cui, C Huang, Y Yang, H Ma).
- Coordinated event planning and recruitment initiatives; recognized as “Excellent Staff of the Month.”

02/2022 - 07/2022 **College of Food Science and Engineering**
Exam Proctor/Grader **Northwest A&F University**

- Graded assignments and exams for Public Speaking (COMM 209).

TALKS & PRESENTATIONS

05/2026 **Nebraska Food Innovation Center**

Tang J. Exploring the Impact of Different Dietary Pulse Market Classes on Human Gut Microbiota and Health. (Public Presentation)

04/2026 **Nebraska Research Days**

Rassmussen C, **Tang J**, Grimaldo C, Benson A, Rose D, Mukhopadhyay I, Majumder K, Deehan E. Effect of Different Nebraska-Dry Bean Market Classes on Human Health and the Gut Microbiota: A Preliminary Analysis. (Poster)

04/2026 **Nebraska Research Days**

Rasmussen C, **Tang J**, Deehan E. Dietary Pulse Market Class Specific Effects on Gut Microbiota, Metabolites, and Human Health: A systematic Review. (Poster)

03/2026 **Missouri Valley Branch ASM Annual Meeting**

Tang J, Grimaldo C, Whisenhunt L, Rasmussen C, Majumder K, Rose D, Deehan E. Effect of Different Nebraska Dry Bean Market Classes on Human Health and the Gut Microbiota: A 3-Phase Randomized Crossover Clinical Trial. (Poster)

04/2025 **Food Science and Technology Symposium**

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

04/2025 **Nebraska Research Days**

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

04/2025 **NPOD 16th Annual Spring Retreat**

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

AWARDS & SCHOLARSHIPS & ASSISTANTSHIPS

2026	The Chancellor's Graduate Research Grant Award	$400
2025 - 2026	Departmental Graduate Research Assistantship	$25500
2025	Excellence in the Department of Food Science and Technology	Certificate
2024	Mary and Charles C. Cooper/Emma I. Sharpless Fellowship	$3000
2024 - 2025	Departmental Graduate Research Assistantship	$24500
2023	National Undergraduate Life Sciences Research - Silver Prize	¥3000
2022 - 2023	Undergraduate Merit Scholarship	¥1500
2021 - 2022	Undergraduate Merit Scholarship	¥1500
2020 - 2021	Undergraduate Merit Scholarship	¥1500

MENTORSHIP

- **Mentor & Co-supervisor,** University of Nebraska-Lincoln (August 2025 – May 2026)

- Mentored and co-supervised one undergraduate research assistant (C Rasmussen) for conducting a systematic review, clinical trial data entry, and research ethics.

- **Mentor,** University of Nebraska-Lincoln (May 2025 – May 2026)
  - Navigated a master student (L Lin) for her academic and social aspects of graduate life and connected her to relevant resources.
- **Mentor,** University of Nebraska-Lincoln (February 2025 – May 2025)
  - Mentored literature search and review for one undergraduate student (Y Yang).
- **Mentor**, Northwest A&F University (January 2023 – June 2023)
  - Mentored in lab techniques for 2 undergraduate students (Y Zhao, J Liu).

LEADERSHIP

09/2025 – 06/2026 **Graduate Student Assembly**
Graduate Representative **University of Nebraska-Lincoln**

- Attended monthly meetings, identifying and advocating on behalf of FDST graduates regarding any issues pertaining to academic affairs.

08/2024 – 08/2025 **Department of Food Science and Technology**
Graduate Student Advisory Board Member **University of Nebraska-Lincoln**

- Organized the "Graduate Student Mentoring Program" to connect each new graduate student with assigned mentors.
- Met with the Departmental Graduate Chair monthly to report on recent questions from current graduate students.

08/2024 – 05/2025 **College of Agricultural Sciences and Natural Resources**
Graduate Representative **University of Nebraska-Lincoln**

- Planned and organized three professional development workshops on professorship, the academic job market, and CV/resume writing for graduate students.

09/2020 – 06/2022 **College of Food Science and Engineering**
Communications Lead **Northwest A&F University**

- Led a team of 13 members to develop promotional newsletters and video content weekly for the College.
- Coordinated with other Colleges' team to co-publish newsletters and video content for the university.

SERVICE

Service to the Departments, Colleges, and Universities

08/2025 **Nebraska Food for Health Center Volunteer**
University of Nebraska-Lincoln

- Hosted an outreach session at the Ice Cream Social event to introduce and advocate for the Nebraska Food for Health Center, engaging students and faculty members in discussions about its research mission and opportunities.

08/2023 – 05/2024 **College of Food Science and Engineering**
Academic Advisor **Northwest A&F University**

- Advised 19 freshmen on their academic planning, research opportunities, and career development.

10/2021 – 08/2023 **College of Food Science and Engineering**
Academic Coordinator **Northwest A&F University**

- Facilitated communication between 59 undergraduate students and the College of Food Science and Technology to address academic and student life concerns.
- Supported "3+1" program events and activities, academic advising sessions, and handled daily student inquiries regarding to their grades and courses.

08/2022 **Office of Undergraduate Studies Volunteer**
Northwest A&F University

- Assisted with student orientation and organized outreach activities to promote student engagement.

Service to Profession

11/2024
Junior Reviewer, *Clinical Nutrition*

Service to Community

06/2022 **College of Food Science and Engineering Volunteer**
Northwest A&F University

- Participated in Undergraduate Rural Voluntary Activities for Poverty Alleviation to deliver outreach and educational programs on food safety for rural school children and families.

PROFESSIONAL DEVELOPMENT

Certifications

- **Python for Data Science, AI & Development**, *Coursera*, *October 2025*
- **Data Science: R Basics**, *HarvardX*, *May 2025*
- **Alberta Food Safety Basics**, *University of Alberta Health Service*, *February 2025*
- **GCP for Clinical Trials with Investigational Drugs and Biologics**, *CITI Program*, *December 2024*
- **Biomedical Responsible Conduct of Research**, *CITI Program*, *January 2024*
- **Conflicts of Interest**, *CITI Program*, *December 2023*
- **Biomedical Investigators and Key Personnel**, *CITI Program*, *December 2023*

Workshops & Trainings

- **Complex Biosystems Seminar Series**, *University of Nebraska-Lincoln. (September 2025 – August 2026)*
- **Nebraska Food for Health Center (NFHC) Seminars**, *Nebraska Food for Health Center. (August 2024 – August 2026)*
- **High Performance Computing for Researchers**, *Holland Computing Center, University of Nebraska-Lincoln. (June 6th – 26th, 2025)*
- **Exploring Your Personal Approach to Leadership, Collaboration, and Well-Being**, *Clifton Strengths Institute. (January 2025 – April 2025)*
- **Professional Food Science Communication**, *Department of Food Science and Technology. (January 2025 – May 2025)*
- **Grant Writing Seminar**, *University of Nebraska-Lincoln. (March 26, 2025)*
- **Food Science and Technology Seminar**, *Department of Food Science and Technology, University of Nebraska-Lincoln. (August 2024 – December 2024)*
- **Peer Mentorship Program**, *Northwest A&F University Psychological Education Association. (January 2021 – February 2021)*

PROFESSIONAL MEMBERSHIPS & AFFILIATIONS

1. ASM Branch Membership-Missouri Valley 2025 – 2026
2. American Society for Nutrition 2024 – 2026
3. American Society for Microbiology 2024 – 2026

4. International Scientific Association of Probiotics and Prebiotics Students and Fellows Association (ISAPP-SFA) 2023 – 2025

SKILLS

- **AI-Agent Computational Biology:** Use AI coding agents (e.g., Claude Code, Codex) to plan, implement, and validate computational workflows for antibody design (framework grafting, structure prediction, structural evaluation) and microbiome analysis
- **Human Clinical Research:** Propose, design, and lead human clinical trials; develop and execute study protocols and SOPs; familiar with IRB/GCP procedures and review requirements; grant proposal writing
- **Wet Lab Techniques:** PCR; blood and stool sample processing (centrifugation, plasma/serum separation, aliquoting, homogenization); −80 °C biobanking; in vitro fermentation; spectrophotometric assays
- **Statistical Analysis:** Experimental design; multivariate analysis and multiple regression, with interpretation of results in biological and clinical context
- **Communication & Languages:** Scientific writing, systematic reviews, and visual communication; Mandarin (native), English (fluent)